Grade 7 Accelerated
Unit 2
Lesson 4 Homework

Name____Answer Key______
Date______________________
Period_____________________

1. Find the square root of each of the following.

$\sqrt{16} =$ 4 $\sqrt{-16} =$ No real solution

2. Find the cube root of each.

$\sqrt[3]{27} =$ 3 $\sqrt[3]{-27} =$ -3

3. Which value from questions 1 and 2 was not a real solution? When does a square or cube root result in a non-real solution?
$\sqrt{-16}$ has no real solution. A square root of any negative number has no real solutions.

Consider $\sqrt{23}$, an irrational number.

4. Between which two perfect squares does 23 lie?
23 is between 16 and 25.

5. Between which two integers is the value of $\sqrt{23}$.
$\sqrt{23}$ is between 4 and 5.

Now consider $\sqrt[3]{55}$, another irrational number.

6. Between which two perfect cubes does 55 lie?
55 is between 27 and 64.

7. Between which two integers is the value of $\sqrt[3]{55}$.
$\sqrt[3]{55}$ is between 3 and 4.

8. Use the approximation of $\sqrt{78}$ to approximate the value of each expression in the table.

Expression	Approximate Value
$\frac{\sqrt{78}}{4}$	$\frac{8.8}{4} = 2.2$
$-3\sqrt{78}$	$-3(8.8) = -26.4$
$\sqrt{78} + 5$	$8.8 + 5 = 13.8$

9. Approximate the value of each expression.

	Expression	Approximate Value
A	$\sqrt[3]{120}$	4.93
B	$\sqrt[3]{120} - 2$	$4.93 - 2 = 2.93$
C	2π	$2(3.14) = 6.28$
D	$1 + \pi$	$1 + 3.14 = 4.14$

Plot and label the approximate location of each expression from question 9 using letters **A**, **B**, **C**, and **D** on the number line.

10.